

THE HEBREW UNIVERSITY OF JERUSALEM

DEPARTMENT OF PHYSICS

Final project

N-body gravitational simulation

Barnes-Hut tree with adaptive RK4 integration

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1 Problem and initial conditions

The simulation follows $N = 5000$ equal-mass particles with total mass $M = 1 \times 10^{11} M_\odot$ for 20 Gyr. Initially, the particles are distributed uniformly within a sphere of radius $R = 50$ kpc. The sampler uses $r = Ru_r^{1/3}$ and chooses an isotropic direction from $\cos \vartheta = 2u_\mu - 1$ and $\varphi = 2\pi u_\varphi$, with u_r , u_μ and u_φ drawn independently from $U(0, 1)$.

Each velocity component follows a normal distribution with $\sigma = V/\sqrt{3}$, so $V = \langle v^2 \rangle^{1/2}$. The three runs use $V = 65, 80$, and 95 km/s. All three reuse the same positions and standardized velocity draw; only the velocity scale changes. The initialization routine subtracts the sampled mean velocity but leaves the positions unchanged, keeping every particle inside the assigned sphere.

The assignment uses the softened force

$$\mathbf{f}_{ij} = -\frac{Gm_i m_j}{(r_{ij} + S)^2} \frac{\mathbf{r}_{ij}}{r_{ij}}, \quad \mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j, \quad (1)$$

with $S = 1$ kpc. Its pair potential, $U_{ij} = -Gm_i m_j / (r_{ij} + S)$, defines the energy diagnostic. After each accepted step, the solver removes particles with any coordinate outside the ± 666 kpc calculation box.

2 Numerical method

2.1 Units

The code uses kpc, $M_\odot$, and Gyr. In these units,

$$G = 4.4996 \times 10^{-6} \text{ kpc}^3 M_\odot^{-1} \text{ Gyr}^{-2}, \quad 1 \text{ kpc/Gyr} = 0.9777922 \text{ km/s}. \quad (2)$$

Each simulation particle has mass $m = M/N = 2 \times 10^7 M_\odot$.

2.2 Barnes-Hut tree

The tree method follows Barnes and Hut [1]. At each force evaluation, the tree builder places the particles in a cubic root cell and splits every cell containing more than one particle into eight octants. Each cell stores its particle count, mass, center of mass, and main diagonal $L = \sqrt{3}l$. The brief defines L as the diagonal, rather than the side length l used in the usual opening criterion.

For each target particle, the tree walk opens any cell that contains it and evaluates a leaf by direct softened interaction, excluding self-interaction. For an external cell at center-of-mass distance D , it recurses if $D/L < \theta$ and replaces the cell by a point mass if $D/L \geq \theta$. The solver uses the assigned $\theta = 1$ and rebuilds the tree at every Runge-Kutta stage.

Dividing Equation 1 by the target mass gives one source's acceleration,

$$\mathbf{a}_{ij} = -\frac{Gm_j}{(r_{ij} + S)^2} \frac{\mathbf{r}_{ij}}{r_{ij}}. \quad (3)$$

Small tests use a direct $O(N^2)$ calculation as the reference. Figure 1 illustrates the same rules with a two-dimensional quadtree. For this 60-particle example, the target requires 12 direct pairs and 14 cell terms instead of 59 direct pairs.

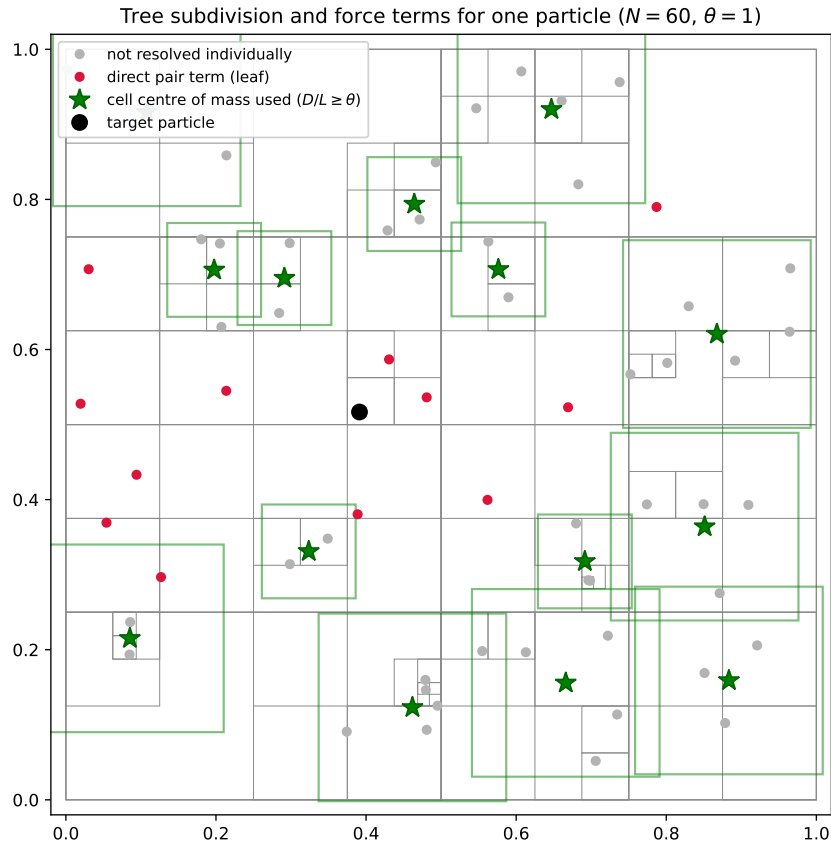


Figure 1: Two-dimensional illustration of the tree method with $\theta = 1$. Thin lines are cell boundaries. Red points contribute a direct pair force to the black target particle. Green stars are cell centers of mass that satisfy $D/L \geq \theta$, drawn inside the cell they replace. Grey points are absorbed into those cells.

2.3 Adaptive RK4 integration

The solver represents the state as $\mathbf{y} = (\mathbf{r}_1, \dots, \mathbf{r}_N, \mathbf{v}_1, \dots, \mathbf{v}_N)$ and integrates

$$\dot{\mathbf{r}}_i = \mathbf{v}_i, \quad \dot{\mathbf{v}}_i = \mathbf{a}_i. \quad (4)$$

For a proposed interval h , the controller compares one fourth-order Runge-Kutta step with two steps of length $h/2$. Their identical first stage is reused, so the comparison costs eleven force evaluations. The brief defines the error as the largest particle displacement difference,

$$\text{err} = \max_i \|\mathbf{r}_{i,\text{half}} - \mathbf{r}_{i,\text{full}}\|. \quad (5)$$

The controller accepts the step when $\text{err} \leq \text{tol}$ and applies Richardson extrapolation,

$$\mathbf{y}_{\text{acc}} = \mathbf{y}_{\text{half}} + \frac{\mathbf{y}_{\text{half}} - \mathbf{y}_{\text{full}}}{2^4 - 1}. \quad (6)$$

The controller multiplies h by $0.9(\text{tol}/\text{err})^{1/5}$, with a maximum growth factor of two and a minimum reduction factor of 0.1. Error control keeps the proposed step between 0.01 and 2 Gyr. Clipping at a reporting time can reduce an accepted interval below this limit. The following one or two intervals may also remain short because h grows by at most a factor of two per accepted step. The shortest interval in the production runs is 0.0016 Gyr. If the error still exceeds the tolerance at the lower bound, the solver stops rather than accept an uncontrolled step.

The brief sets the displacement accuracy to $\varepsilon = 0.1$ in units of the box size, yielding

$$\text{tol} = \varepsilon L_{\text{box}} = 0.1 \times 1332 \text{ kpc} = 133.2 \text{ kpc}, \quad (7)$$

a permitted step error of 133 kpc for an initial cluster radius of 50 kpc. Section 4 tests this value.

The integration begins at 0 Gyr and clips steps to land exactly at 2.5, 5, 10, 15, and 20 Gyr, so each stored snapshot matches its label.

2.4 Energy and escapers

At every accepted step, the energy diagnostic calculates

$$E_k = \frac{1}{2} \sum_i m_i v_i^2, \quad E_g = \frac{1}{2} \sum_i m_i \Phi_i, \quad \Phi_i = -G \sum_{j \neq i} \frac{m_j}{r_{ij} + S}. \quad (8)$$

For $N > 1500$, the diagnostic evaluates Φ_i with the force tree; smaller tests use the direct sum. When particles leave, the code stores their kinetic energy and estimates E_g before and after removal. Their difference includes escaper-escaper and escaper-retained terms within the chosen estimator. At production size, both estimates use separate Barnes-Hut approximations. The retained curve includes only active particles; the corrected curve adds the frozen escape contributions. Neither includes later interactions because removed particles no longer evolve. The removal rule is geometric, so escaper here means boundary-removed rather than proven unbound: at $V = 80$, 21 of the 123 removal events left with non-positive $E_k + E_g$.

2.5 Density profile and King fit

Radii are measured from the retained center of mass, which is displaced by 2 to 8 kpc after escaper removal. The occupied radial range is divided into 25 logarithmic shells. For shell edges r_j and r_{j+1} ,

$$\rho_j = \frac{M_j}{\frac{4\pi}{3}(r_{j+1}^3 - r_j^3)}, \quad r_{j,\text{plot}} = \sqrt{r_j r_{j+1}}. \quad (9)$$

For equal mass particles, the Poisson uncertainty is $\delta\rho_j = \rho_j/\sqrt{N_j}$. The implementation also supports unequal masses by summing m_i^2 in each shell.

The fit uses the model requested in the assignment,

$$\rho(r) = \frac{\rho_c}{[1 + (r/r_c)^\alpha]^\beta}. \quad (10)$$

The analysis fits $\log \rho$ with propagated error $\delta \log \rho \simeq \delta \rho / \rho$. Optimizing $\log \rho_c$ and $\log r_c$ keeps both parameters positive. The $V = 80$ fit scans several starting values for α and β , then holds them fixed while fitting ρ_c and r_c for $V = 65$ and $V = 95$.

3 Implementation and checks

The submitted `nbody_solver.py` implements sampling, tree forces, adaptive integration, diagnostics, fitting, plotting, and self-tests. Numba compiles the flattened tree and distributes target particles across threads; the Python walk is the fallback. The RK controller from HW_05 uses the per-particle norm in Equation 5 and resizes the state after escapes. The isotropic direction sampler comes from HW_07.

The command `python nbody_solver.py --test` runs the checks, and `--tol=X` runs all three $N = 5000$ cases. Outputs are stored under `figures/`; redraw and refit modes reuse saved states without repeating the dynamics.

In the self-test, 5000 samples give a maximum radius of 49.9905 kpc and an RMS speed of 79.422 km/s for the 80 km/s distribution. The production draw gives 49.9961 kpc and 80.235 km/s. These sub-percent deviations match sampling noise, and subtracting the mean sets the center-of-mass velocity to machine zero.

For 150 particles with unequal masses, the Barnes-Hut acceleration at $\theta = 1$ had a relative error $\|\mathbf{a}_{\text{tree}} - \mathbf{a}_{\text{dir}}\| / \|\mathbf{a}_{\text{dir}}\| = 1.014 \times 10^{-2}$ against the direct sum, with both norms stacked over all particles and components. The tree gravitational energy differed from the direct result by 5.214×10^{-4} . A softened unequal-mass circular binary completed two periods with a relative energy change of 1.29×10^{-10} under a tight tolerance, and its final separation agreed with the initial value to the reported precision. A separate constant-velocity test confirmed that stored snapshots match their labels.

The brief's opening convention differs from the standard side-length form. Since $L = \sqrt{3}l$,

$$\frac{D}{L} \geq \theta_{\text{diagonal}} \iff \frac{l}{D} \leq \frac{1}{\sqrt{3}\theta_{\text{diagonal}}} \equiv \theta_{\text{side}}. \quad (11)$$

Table 1 compares the tree acceleration with a direct sum for 3000 particles. At the assigned $\theta = 1$, the L_2 error is 9.14×10^{-3} , consistent with the smaller random test.

Table 1: Barnes-Hut acceleration error against direct summation for the initial uniform sphere, $N = 3000$.

θ (diagonal)	θ (side)	L_2 relative error in a
2.0	0.289	1.17e-03
1.5	0.385	3.07e-03
1.0	0.577	9.14e-03
0.8	0.722	1.58e-02
0.6	0.962	3.55e-02
0.4	1.443	1.03e-01

Raising θ to 2 cuts the error by a factor of eight. At $N = 5000$, tree construction dominates the runtime, so changing θ has little effect. The production runs use the assigned $\theta = 1$. The production controller accepted 310, 262 and 352 steps for $V = 80, 95$ and 65 km/s, with 4, 10 and 3 rejected proposals. The assigned tolerance produced no rejections in the corresponding full runs.

4 Choice of integration tolerance

The assigned tolerance, 133.2 kpc, exceeds the initial cluster radius. The tolerance study repeats the $V = 80$ km/s run at four values across three decades. For each run, the energy diagnostic measures the retained particle energy plus the escape values frozen at removal. Adding these values removes deletion jumps but does not make the diagnostic exact: the Barnes-Hut force and tree energy both introduce spatial approximations. Comparing the runs measures sensitivity to timestep error while these approximations remain at every tolerance.

Table 2: Energy drift versus RK4 position tolerance for $V = 80$ km/s, $N = 5000$, 20 Gyr, including frozen escape values.

tol [kpc]	steps	rejected	N_f	$\max \Delta E/E_0 $
133.2	20	1	4781	4.82e-01
13.32	33	0	4833	8.37e-02
1.332	97	0	4829	1.77e-02
0.1332	303	4	4833	2.53e-03

Tightening the tolerance to 0.1332 kpc lowers the worst energy excursion from 48 percent to 0.25 percent while increasing the step count from 20 to 303. The main results use this value; Appendix A provides every required output at the assigned tolerance under the suffix `_eps01`.

The drift falls at every row of Table 2, so the study does not reach a floor. Further tightening would eventually expose timestep-independent tree and close-encounter errors, but these runs do not locate that crossover.

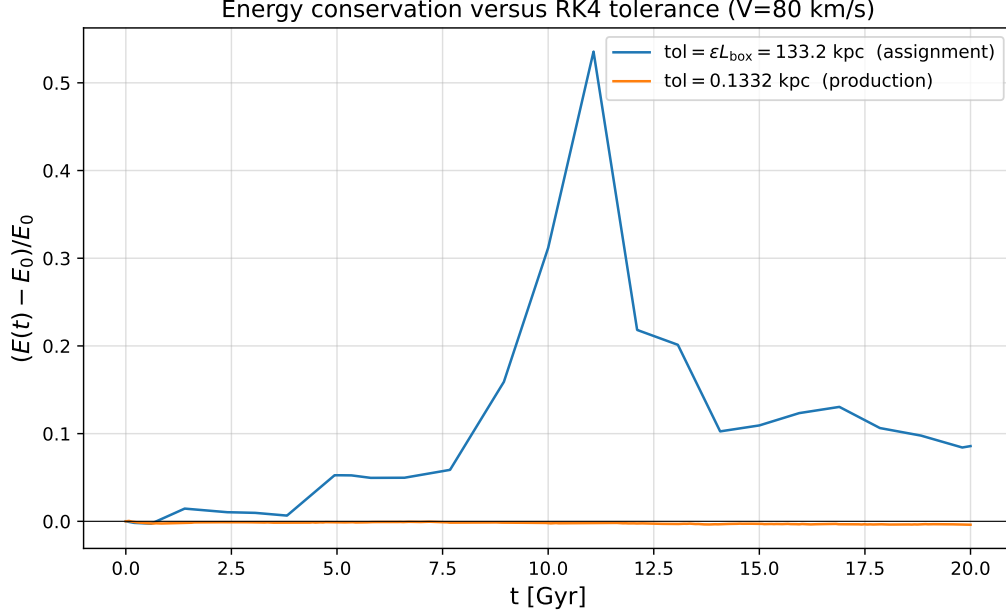


Figure 2: Relative energy change for $V = 80$ km/s at the tolerance of the assignment and at the tolerance used for the production runs.

5 Results

5.1 Initial energies

For an unsoftened uniform sphere, $E_g = -3GM^2/(5R)$. Since $E_k = MV^2/2$, the virial value $2E_k/|E_g| = 1$ occurs at

$$V_{\text{vir}} = \sqrt{\frac{3GM}{5R}} = 71.85 \text{ km/s}. \quad (12)$$

The $V = 65$ case starts below virial equilibrium, while the $V = 80$ and $V = 95$ cases start above it. All three have $2E_k/|E_g| < 2$ and negative initial total energy. The $V = 95$ state is close to zero energy, but it is not initially unbound.

Table 3: Initial and final run data. Energies use the simulation units.

V [km/s]	$2E_{k,0}/ E_{g,0} $	E_0 [$10^{14} M_\odot \text{ kpc}^2 \text{ Gyr}^{-2}$]	removed	retained	steps
65	0.849	-3.013	20	4980	352
80	1.286	-1.868	167	4833	310
95	1.814	-0.488	758	4242	262

5.2 (a) Particle positions

Figure 3 shows the $V = 80$ positions at the six requested times. The panels frame the cluster because the full box would hide its structure. By 2.5 Gyr, the uniform sphere has formed a dense core and an extended halo. The halo expands throughout the run, and 167 particles cross the box boundary.

5.3 (b) Energy conservation

The curves in Figure 4 separate after the first escape at 5.3 Gyr. Since $E_0 < 0$, a positive value of $(E - E_0)/E_0$ means the total energy has become more negative. At 20 Gyr, the retained energy gives +0.034 and the frozen-escape accounting gives -0.0039 . The removed particles carry away kinetic energy equal to 4.6 percent of $|E_0|$, but only 0.8 percent in negative interaction energy. The retained total therefore becomes more negative and its relative-change curve rises. At $V = 95$, where 758 particles escape, the two final values are +0.81 and +0.0013; the corrected curve peaks at 0.0069 earlier in the run. At $V = 65$, only 20 leave; the retained and corrected values end at -0.0020 and -0.0032 .

5.4 (c) Virial ratio

The $V = 80$ ratio starts at 1.286, falls to 0.82 by 2.5 Gyr, and then oscillates about one with decreasing amplitude (Figure 5). Maxima at 5.1, 9.4, 14.0 and 18.6 Gyr give a 4.5 Gyr period. The final retained value is 0.954. The damped oscillation toward unity is consistent with violent relaxation.

The final retained ratios are 0.960, 0.954 and 0.937 for $V = 65$, 80 and 95 km/s. The $V = 65$ system starts at 0.849, contracts, and loses 20 particles. The $V = 95$ system starts at 1.814 and loses 15 percent of its mass as it approaches equilibrium. Its two escape-accounting curves differ most.

The plotted ratio $-2E_k/E_g$ is the virial condition requested in the brief, which holds exactly for an unsoftened potential. Since $U_{ij} \propto (r_{ij} + S)^{-1}$ is not homogeneous of degree -1 , the softening-consistent statement is $2E_k + W = 0$ with $W = \sum_{i<j} \mathbf{r}_{ij} \cdot \mathbf{f}_{ij} = -\sum_{i<j} Gm_i m_j r_{ij} / (r_{ij} + S)^2$. Summing W directly over the final states gives $2E_k/|W| = 0.9996, 0.982$ and 0.956 for $V = 65, 80$ and 95 km/s, closer to unity than the plotted values in all three runs.

5.5 (d) Density profiles and King fits

The fitted core radius increases monotonically with initial velocity, from 24.7 kpc at $V = 65$ through 35.1 kpc at $V = 80$ to 57.4 kpc at $V = 95$. The central density falls by more than a factor of ten over the same range, from 7.5×10^5 to $5.4 \times 10^4 M_\odot \text{ kpc}^{-3}$. Higher initial velocity leaves a larger, more diffuse remnant (Figure 6).

Table 4: King-model parameters. The uncertainties are the formal one-standard-deviation errors from the weighted fit. For $V = 65$ and 95 they are conditional on the α and β held fixed from $V = 80$.

V [km/s]	$\rho_c [M_\odot/\text{kpc}^3]$	$r_c [\text{kpc}]$	α	β
80	$2.455\text{e}+05 \pm 1.09\text{e}+04$	35.125 ± 0.447	4.831 ± 0.385	0.986 ± 0.085
95	$5.380\text{e}+04 \pm 1.59\text{e}+03$	57.339 ± 0.506	4.831 (fixed)	0.986 (fixed)
65	$7.469\text{e}+05 \pm 2.16\text{e}+04$	24.673 ± 0.211	4.831 (fixed)	0.986 (fixed)

The common shape parameters are $\alpha = 4.83 \pm 0.38$ and $\beta = 0.986 \pm 0.085$. The outer profile constrains their product because $\rho \propto r^{-\alpha\beta}$ at large radius. Here $\alpha\beta = 4.77$, close to the Plummer value of 5. The assigned-tolerance states give $\alpha = 0.700$, $\beta = 7.389$, and a

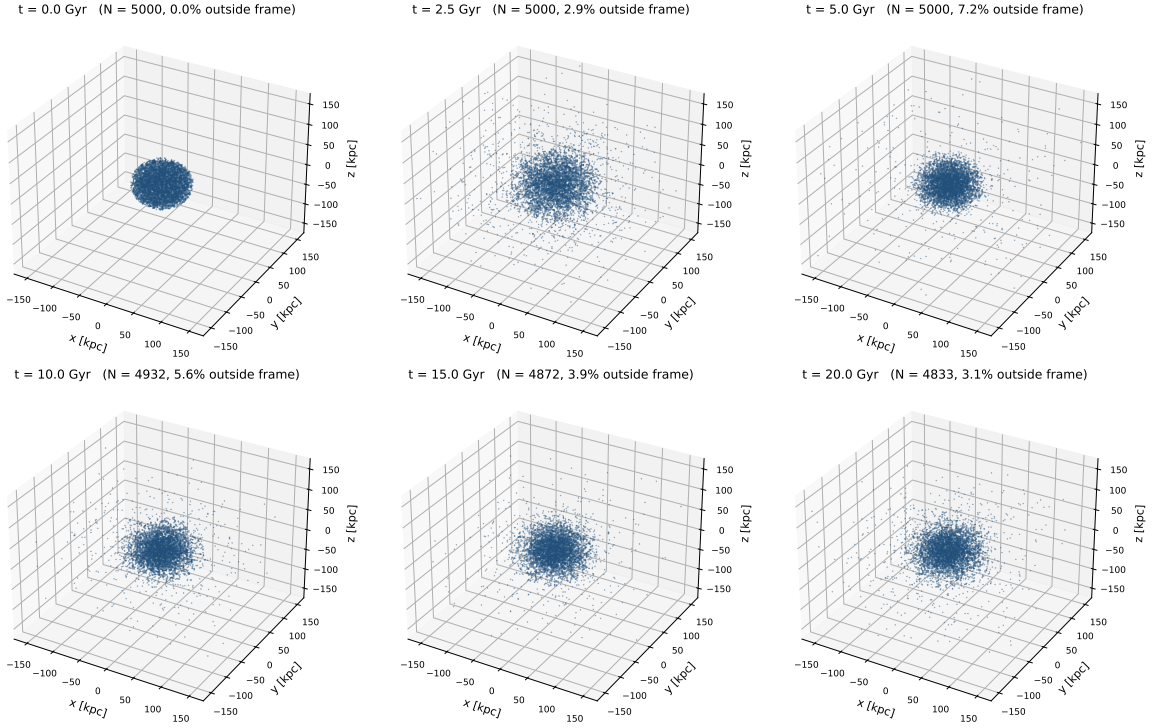


Figure 3: Particle positions for the $V = 80$ km/s run. Each panel shows an integrated state at the time printed above it.

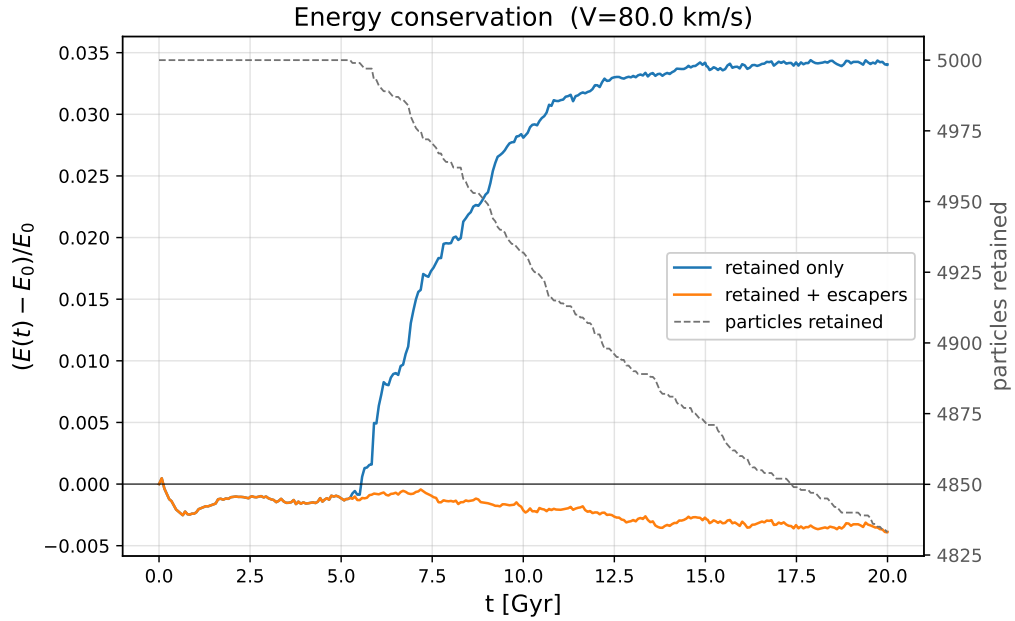


Figure 4: Relative energy change for $V = 80$ km/s. Blue uses retained particles, orange restores each escaper's energy recorded at removal, and the dashed grey line gives the retained count on the right axis.

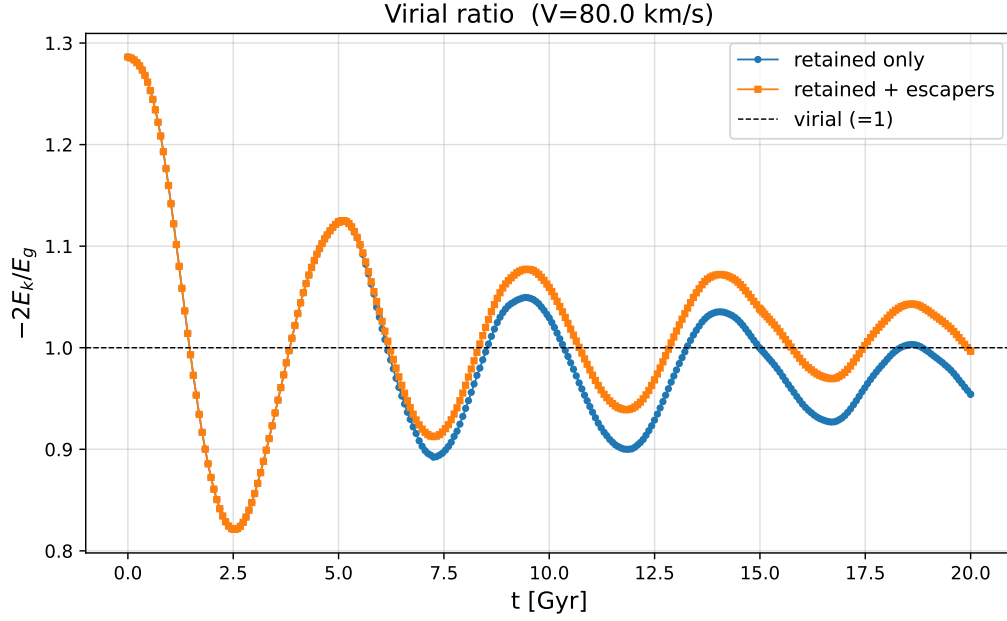


Figure 5: Virial ratio for $V = 80$ km/s. The horizontal line marks $-2E_k/E_g = 1$.

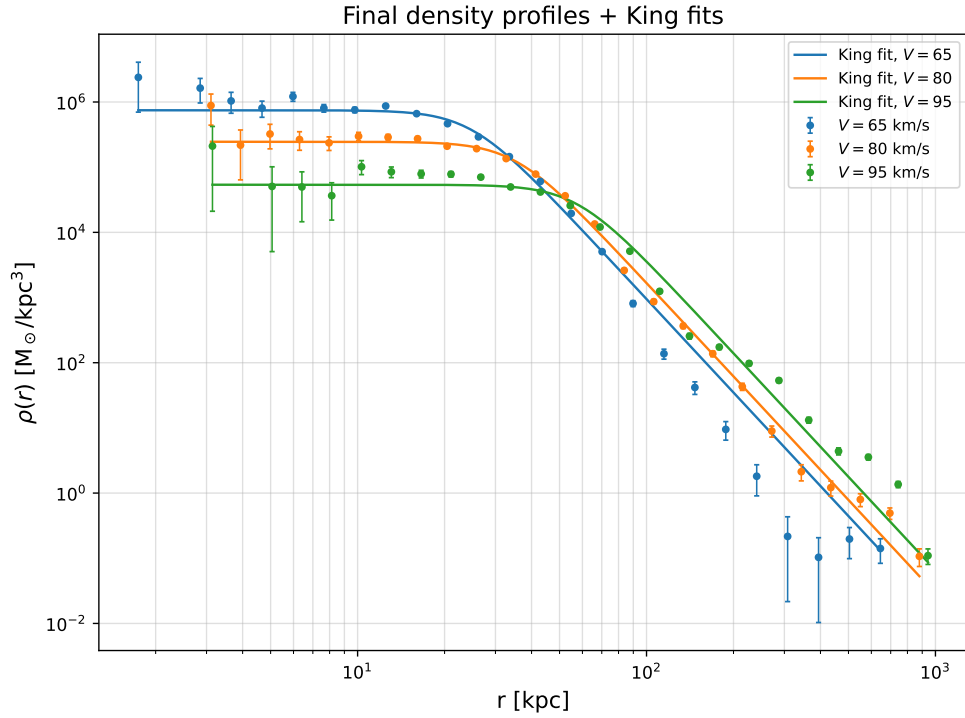


Figure 6: Center-of-mass density profiles at 20 Gyr with King-model fits. The $V = 65$ and $V = 95$ fits use the α and β values obtained from $V = 80$.

similar product of 5.17. The data therefore constrain the outer slope more strongly than either shape parameter.

The reduced χ^2 values, 7.7, 44.7 and 12.8 for $V = 80, 95$ and 65 km/s, exceed unity. Populated bins have small Poisson errors, so departures from the King form dominate the statistic; the $V = 95$ profile also has a shallow central depression that a monotonic King curve cannot fit. The quoted uncertainties describe local parameter curvature, not model mismatch.

6 Numerical difficulties

The assigned tolerance exceeds the initial cluster diameter and permits energy errors of tens of percent (Section 4). RK4 is non-symplectic and therefore lacks the bounded long-time energy error typical of leapfrog methods. The drift decreases at every tested tolerance. The Barnes-Hut force and energy approximations do not depend on the timestep, and these runs do not locate the tree- and encounter-dominated error floor.

Close encounters raise the acceleration and can drive the step to its 0.01 Gyr floor. Softening bounds the pair force by $Gm_i m_j / S^2$; if the error remains too large at the floor, the code stops rather than accepting an uncontrolled step. Exact coordinate duplicates require a separate tree-build guard because repeated octant subdivision cannot separate them. The tree stores such particles in one childless cell. The group acts as a single point mass for an external target, which is exact. Their mutual force is set to zero, which is a regularization rather than an evaluation of Equation 1: at $r_{ij} = 0$ that law has finite magnitude $Gm_i m_j / S^2$ and undefined direction. The chosen initial sampler does not generate duplicates.

Computing E_g with the $\theta = 1$ tree walk costs $O(N \log N)$ at every accepted step. Its relative bias is 5.2×10^{-4} against the direct sum for 150 particles, below the measured drift; runs below 1500 particles use the exact sum. Escaper removal causes jumps in the retained-particle energy by deleting both the particle’s kinetic energy and its pair terms. The corrected curves in Figures 4 and 5 remove these jumps from the plotted diagnostics, but retain tree-estimator error and omit post-removal interactions.

For the density fit, 25 logarithmic shells resolve the core better than a linear grid; empty bins are omitted and the remaining points use Poisson weights. Because the outer model mainly constrains $\alpha\beta$, the log-space fit tries several starting points and retains the lowest χ^2 . Fixing the $V = 80$ shape for the other two velocities reduces this degeneracy and satisfies the assignment.

Adaptive step doubling needs eleven force evaluations per attempted step because the full step and first half step reuse their identical first stage. Each remaining force evaluation builds a new tree. Flattened tree arrays, a Numba-compiled walk and threaded target-particle loops make the $N = 5000$ calculation tractable.

7 Conclusion

Direct-sum tests validate the tree calculation; softened-binary and snapshot tests validate the integrator. After 20 Gyr, all three systems approach $-2E_k/E_g \approx 1$: the 65 km/s system contracts and retains almost all its mass, while the 80 km/s and 95 km/s systems expand and lose about 3 and 15 percent. Their fitted core radii increase and central densities decrease with initial speed.

At the assigned 133.2 kpc tolerance, final escape-corrected energy shifts range from 9 to 35 percent. Reducing it by 10^3 brings all three final shifts below 0.4 percent, and the largest excursion at any time below 0.7 percent, at the cost of 310 rather than 28 accepted steps for the full-output 80 km/s case. The main results therefore use the tighter tolerance; Appendix A retains every requested output at the assigned value.

A Results at the assigned tolerance

This appendix repeats all three runs with the assigned $\text{tol} = \varepsilon L_{\text{box}} = 133.2$ kpc instead of the tighter 0.1332 kpc. The files use the `_eps01` suffix and the command

```
python nbody_solver.py --tol=133.2 --tag=eps01.
```

The initial conditions, θ , S , box, and end time are unchanged.

The controller accepts 28, 15 and 34 steps for $V = 80, 95$ and 65 km/s, with no rejections. The tolerance study in Table 2 uses a smaller step floor and no intermediate snapshot times, so its accepted sequence and drift differ slightly. In both configurations, the tolerance exceeds the initial cluster diameter and provides little step control.

Table 5: Assigned and tighter runs. $\Delta E/E_0$ is the final value of the escaper-corrected relative energy change, not its largest excursion; the assigned-tolerance $V = 80$ run peaks at 0.54 and the tighter $V = 95$ run at 0.007.

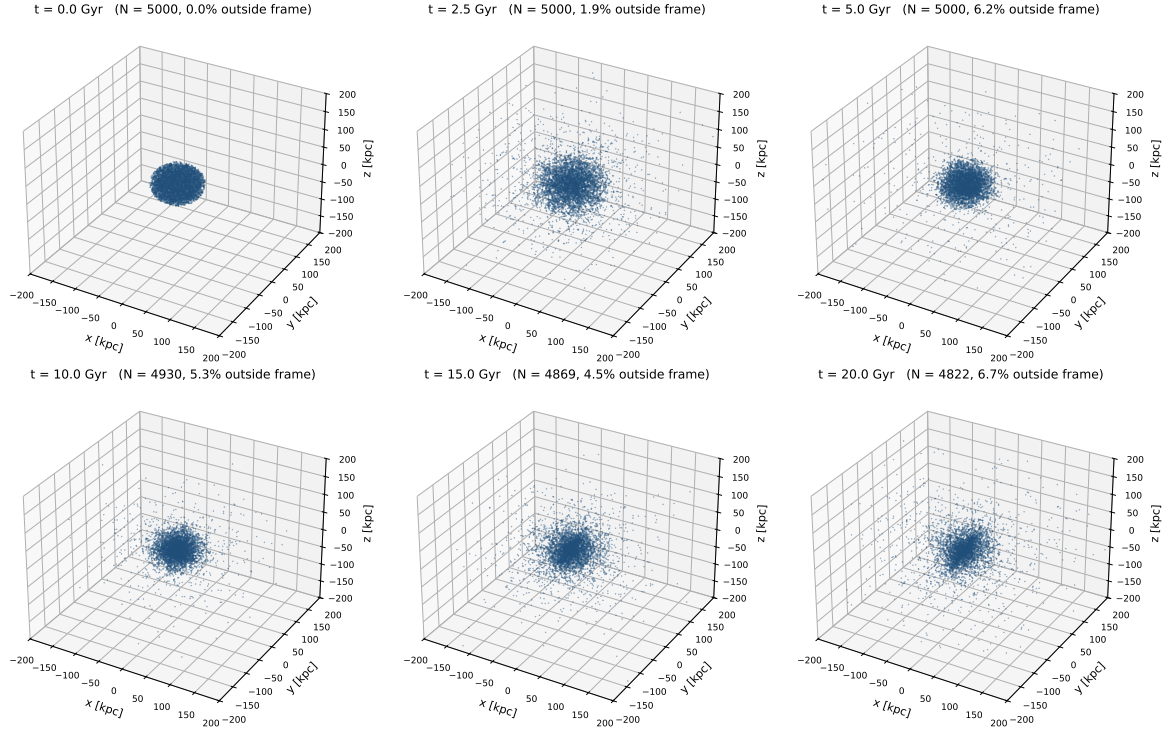
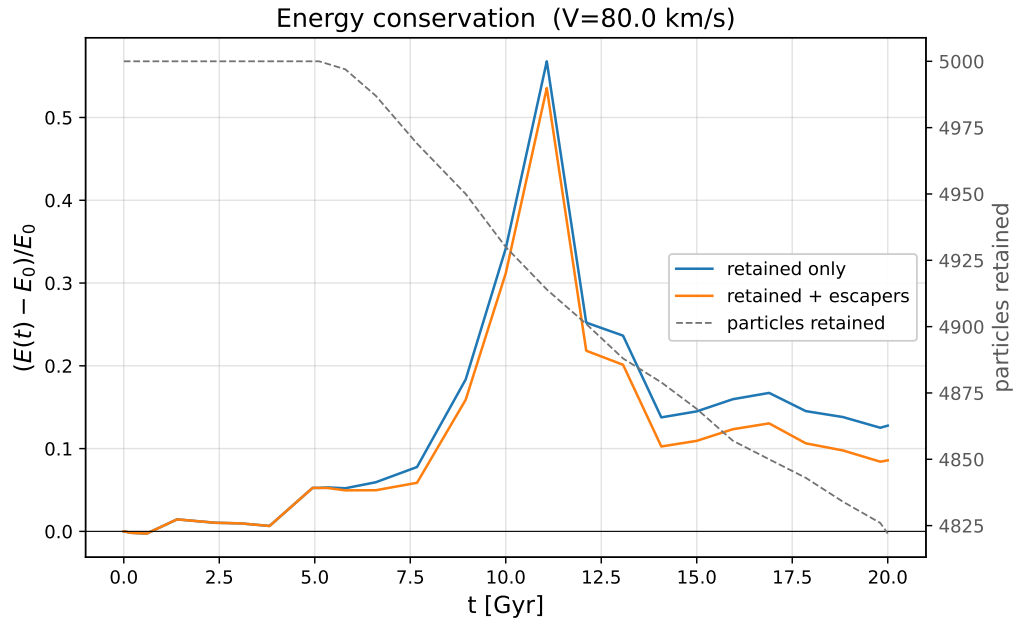
V [km/s]	tol = 133.2 kpc (assigned)			tol = 0.1332 kpc (used)		
	steps	retained	$\Delta E/E_0$	steps	retained	$\Delta E/E_0$
65	34	4755	−0.347	352	4980	−0.003
80	28	4822	+0.086	310	4833	−0.004
95	15	4239	+0.219	262	4242	+0.001

Table 6: Assigned-tolerance King parameters. The $V = 80$ fit supplies α and β for the other velocities.

V [km/s]	ρ_c [$M_\odot/\text{kpc}^3$]	r_c [kpc]	α	β
80	$2.642\text{e}+07 \pm 6.36\text{e}+06$	25.217 ± 3.541	0.700 ± 0.050	7.389 ± 0.829
95	$6.504\text{e}+06 \pm 4.26\text{e}+05$	39.954 ± 0.843	0.700 (fixed)	7.389 (fixed)
65	$2.872\text{e}+07 \pm 1.33\text{e}+06$	24.525 ± 0.363	0.700 (fixed)	7.389 (fixed)

At the assigned tolerance, the corrected energy error reaches 54 percent in the full-output $V = 80$ run. Table 2 gives 48 percent because a different step floor and output schedule alter the accepted sequence. The virial ratio ends near unity, 0.929 rather than 0.954, but its 28 samples cannot resolve the 4.5 Gyr period.

The density fits are less reliable. Table 6 gives $r_c = 24.5, 25.2$ and 40.0 kpc for $V = 65, 80$ and 95 km/s; the first two agree within error, erasing the tighter-run trend. At 65 km/s, the coarse cusp raises the central density from 7.5×10^5 to $2.9 \times 10^7 M_\odot \text{kpc}^{-3}$. Although (α, β) changes from (4.831, 0.986) to (0.700, 7.389), their product remains close, 4.77 versus 5.17. The loose run removes 245 particles instead of 20.

Figure 7: Assigned-tolerance particle positions for $V = 80$ km/s, requirement (a).Figure 8: Assigned-tolerance energy and population for $V = 80$ km/s, requirement (b).

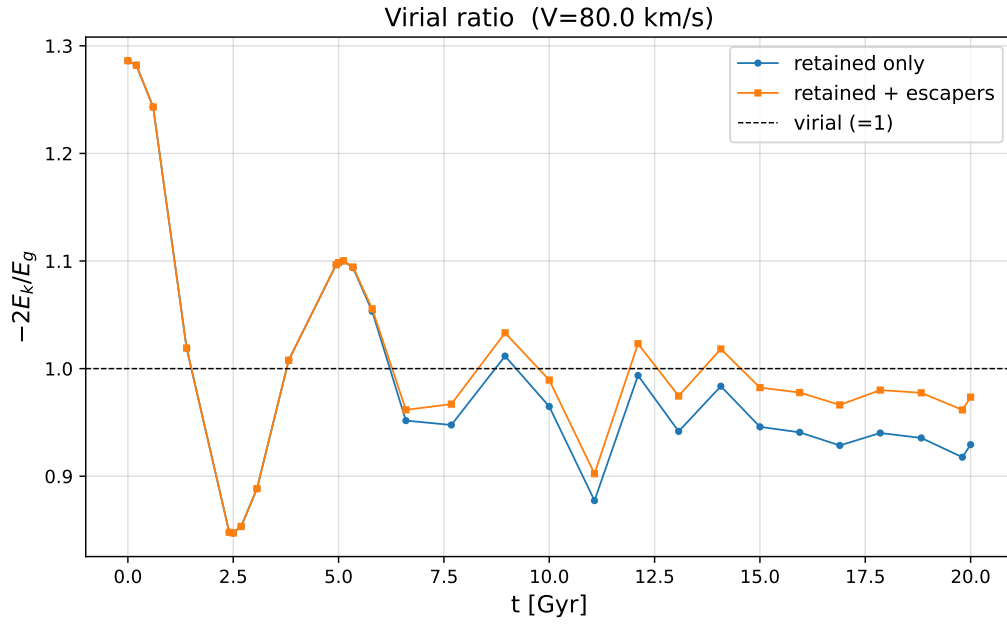


Figure 9: Assigned-tolerance virial ratio for $V = 80$ km/s, requirement (c).

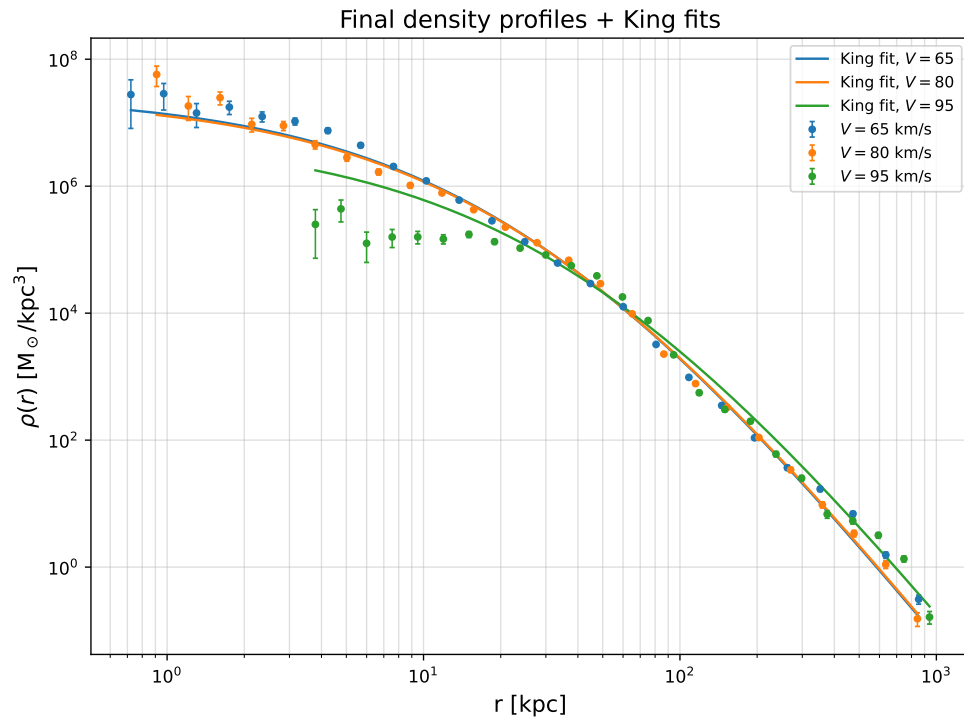


Figure 10: Assigned-tolerance density profiles and King fits, requirement (d).

References

- [1] J. Barnes and P. Hut, “A hierarchical $O(N \log N)$ force-calculation algorithm,” *Nature* **324**, 446–449 (1986).